



Visual Grounding for Satellite Images

Dual Degree Project Stage II

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Introduction

Visual Grounding: The task of localizing a specific object or region in an image based on a natural language description.

Applications

- Satellite Image Analysis
- Robotics and Autonomous Systems
- Human-Computer Interaction
- Image Retrieval and Search

Challenges

- Complex and Cluttered Scenes
- Ambiguous Language Descriptions
- Domain-Specific Knowledge
- Scalability and Efficiency



Single-stage vs. Two-stage Approaches

- Two-stage approaches:
 - First generate a set of sparse region proposals and then exploit region-expression matching to find the best one
 - Either binary classification loss or maximum-margin ranking
 - High accuracy but computationally expensive
- One-stage approaches:
 - Perform visual-linguistic fusion at intermediate layers of an object detector and output the box with maximal score over pre-defined dense anchors
 - Linguistic context is densely fused with the visual features, bounding box prediction in a sliding-window manner
 - Faster and Simpler architecture, with potentially lower accuracy



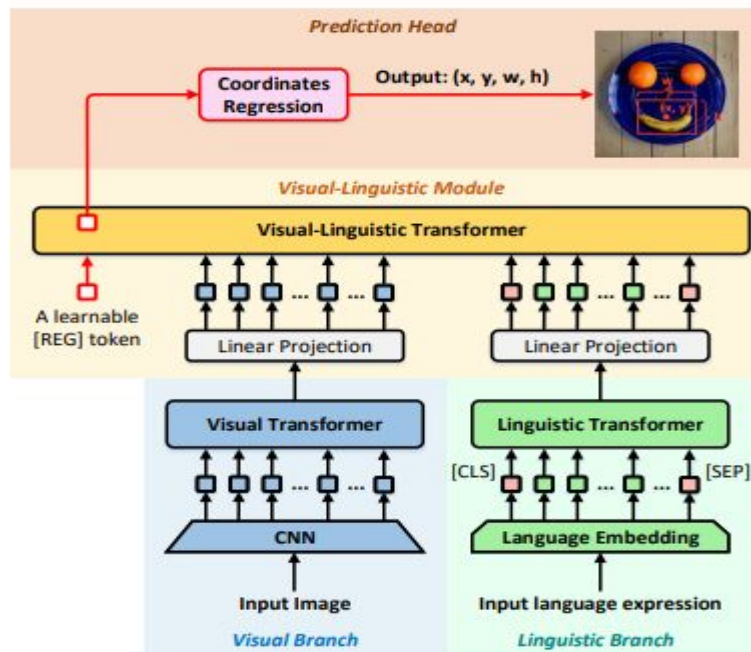
Transformer-based Approaches

TransVG

- Reformulate the visual grounding as a direct coordinates regression problem and avoid masking predictions out of a set of candidates, outputs a 4-dim coordinates to ground the object
- Separate encoding for video and linguistic features, then append a multi-modal learnable token [REG] for Visual-Linguistic Transformer
- The appended token is leveraged to directly predict the 4 dim coordinates of the referred object
- Visual Branch: ResNet is used as the convolutional backbone network, and initialized with the encoder of DETR model with 6 transformer encoder layers, with 8 heads in multi-head self-attention
- Linguistic Branch: BERT is used to generate the features and a similar transformer structure as the visual branch
- Visual-Linguistic Fusion Module: The visual and linguistic tokens are concatenated and appended with learnable token

Experimental setup, loss, datasets, results

- ReferItGame: 20k images with each image having few regions with corresponding referring expressions (54k)
- Flickr30K: 31k images with 427k referred entities
- RefCOCO: 20k images with 50k referred objects.
- Smooth L1 loss, with normalized coordinates
- IoU Loss





Visual Grounding with Transformers

- Approach is to learn semantic-discriminative visual features under the guidance of the textual descriptions without harming their location ability - reformulates visual grounding as the problem of object bounding box coordinates regression conditioned on the query sentence.
- Basic Visual and Textual Encoders: ResNet to extract image features and RNN based soft-parser to extract textual tokens (bi-directional LSTM is applied to encode the context for each word)
- Grounding Encoder:
 - Textual branch with self-attention
 - Visual Branch with Text-guided Self-attention
- Grounding Decoder: Modified textual tokens serve as the grounding queries to decode the text-guided visual features
- Prediction Head: Used for bounding box coordinate regression

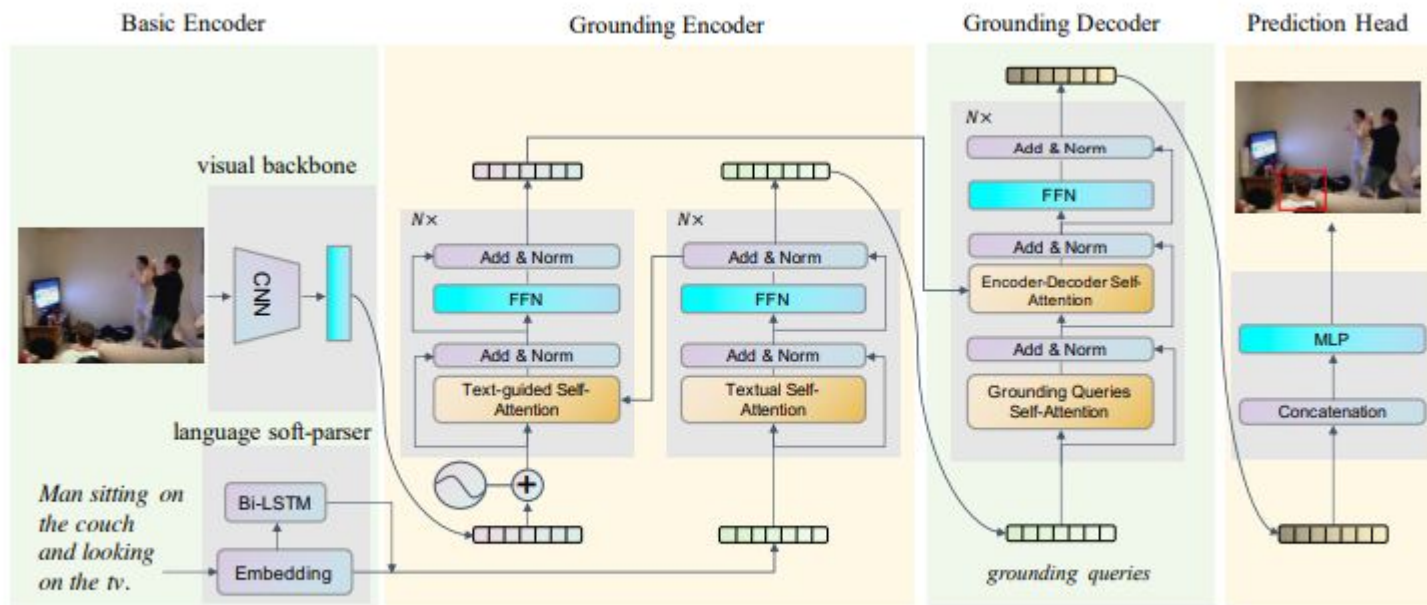


Fig. 1. The overall architecture of the VGTR framework.



Limitations of Existing Methods

- Limited performance on complex, cluttered scenes
- Struggle with domain-specific terminology
- Computationally intensive for high-resolution images
- Lack of focus on satellite imagery domain
- Difficulty in handling multi-object relationships
- Limited interpretability of decisions
- Challenges in scaling to large datasets



Dataset Overview

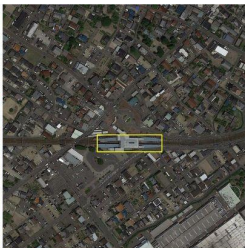
Dataset	Images	Referred Entities	Domain
ReferItGame	20,000	54,000	General
Flickr30K	31,000	427,000	General
RefCOCO	20,000	50,000	General
RSVG	10,000	30,000	Satellite

RSVG dataset focuses on diverse geographical high-resolution satellite images with domain specific language description

Sample Images for RSVG Dataset



A green and brown golf field



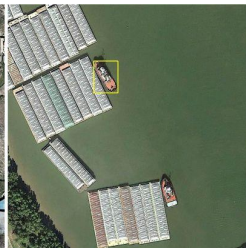
The small slender train station in the middle



The chimney in the left



A white storage tank in the middle



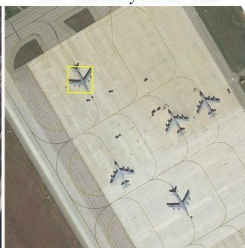
The ship on the top



A vehicle in the middle



The long large dam



A airplane on the upper left



The oval large green and red ground track field



A gray stadium in the middle



Proposed Single-stage approach

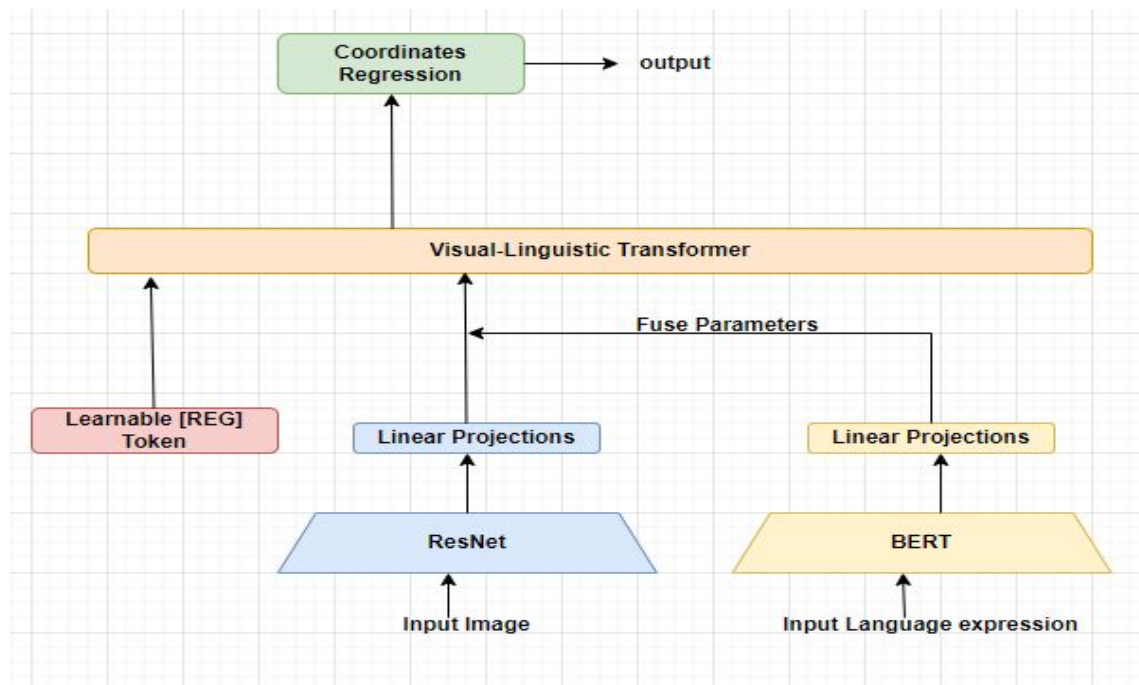
Key Components:

- ResNet for Visual Encoding
- BERT for Linguistic Encoding
- Cross-Modal Attention Mechanism
- Bounding Box Prediction Module

Innovations:

- Adapted ResNet for high-resolution satellite imagery
- Domain-specific pre-training on satellite image captions
- Multi-scale feature fusion for improved spatial understanding
- Attention-guided bounding box refinement

Model Architecture





Training and Loss

Pipeline

- Data preprocessing and augmentation
- Model Initialization
- Iterative training with multi-task loss
- Validation and early stopping

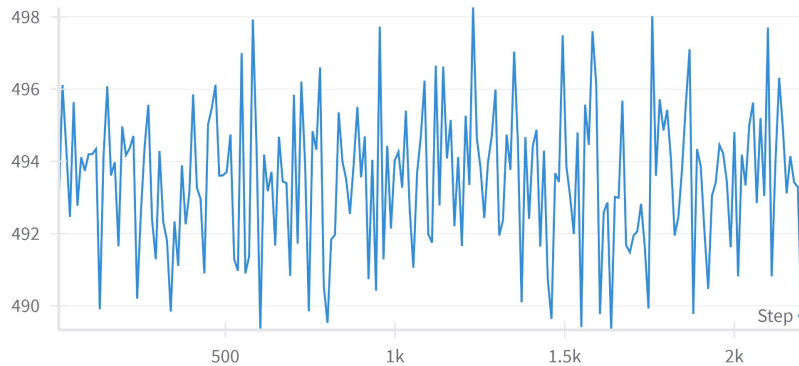
Loss Functions

- IoU Loss
- Smooth L1 Loss for bounding box regression

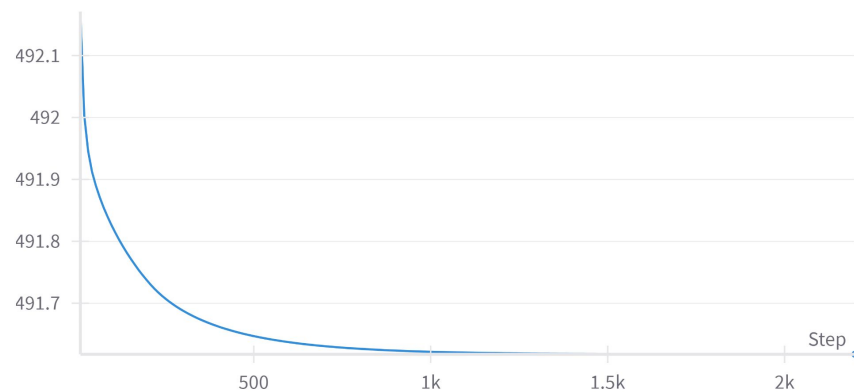


Training and Validation Curves

avg_train_loss



val_loss





Results and Metrics

Model	Accuracy	IoU
TransVG (Natural)	71.85%	0.621
TransVG (Satellite)	33%	0.235



Future work

- Replace ResNet with Vision Transformer (ViT)
- Fine-tune both the Visual and Linguistic encoders on satellite data
- Enhance multi-modal attention mechanisms
- Increase dataset size and diversity



References

<https://github.com/djiajunustc/TransVG>

<https://github.com/ZhanYang-nwpu/RSVG-pytorch>

<https://paperswithcode.com/paper/towards-vision-language-geo-foundation-model>

<https://paperswithcode.com/paper/skyeyegpt-unifyin-g-remote-sensing-vision>

<https://paperswithcode.com/paper/earthmarker-a-visual-prompt-learning>

<https://dl.acm.org/doi/abs/10.1145/3503161.3548316>

<https://arxiv.org/pdf/1608.00272>

<https://arxiv.org/abs/1505.04870>

<https://aclanthology.org/D14-1086/>

<https://arxiv.org/abs/2104.08541>

Slide 1: Title Slide

- **Title:** "SatVG: Advanced Vision-Language Grounding for Satellite Imagery"
- **Subtitle:** "A Comprehensive Study of Transformer-based Architectures for Remote Sensing Visual Understanding"
- **Visual:** Custom diagram showing satellite imagery with bounding boxes and text queries

Slide 2: Research Motivation

- Remote sensing analysis requires expert knowledge
- Natural language interfaces can democratize satellite imagery analysis
- Visual grounding bridges the gap between language and visual understanding
- **Visual:** Split-screen showing traditional interfaces vs. natural language query example

Slide 3: Technical Challenges

- Satellite imagery has unique characteristics (scale variation, orientation ambiguity)
- Limited annotated datasets for satellite visual grounding
- Complex spatial relationships difficult to capture with traditional architectures
- Cross-modal alignment between specialized satellite terminology and visual features
- **Visual:** Examples of challenging satellite image queries with varying scales

Slide 4: Dataset Analysis - Overview

- Comprehensive analysis of RSVG dataset
- 10,665 total samples with 9,275 valid samples
- Mean box size 16.6% vs median 4.5% (significant small object bias)
- Average text query length: 10.1 words
- **Visual:** Include dataset-analysis/box_size_distribution.png

Slide 5: Dataset Analysis - Deep Dive

- Right-skewed distribution with many small objects
- Slight preference for wider objects (mean ratio 1.17)
- Significant portion (13%) with missing images required preprocessing
- Detailed word frequency analysis revealed domain-specific terminology
- **Visual:** Include dataset-analysis/box_ratio_distribution.png and dataset-analysis/query_word_frequencies.png

Slide 5: Dataset Analysis - Deep Dive

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Slide 6: Architectural Innovation

- Multi-phase research approach spanning traditional to cutting-edge architectures
- Designed novel architectural variants for remote sensing specificity
- Developed specialized fusion modules for satellite domain terminology
- Integrated position-aware attention mechanisms for improved spatial understanding
- **Visual:** Include satvg-architecture.png

Slide 7: Backbone Exploration - Phase 1

- Systematic evaluation of CNN vs Transformer architectures
- ResNet50 backbone established baseline performance (67.09% accuracy)
- Initial identification of localization challenges in satellite imagery
- Critical insights into feature map resolution importance
- **Visual:** Architecture diagram highlighting ResNet components

Slide 8: Backbone Exploration - Phase 2

- Vision Transformer integration required significant architectural adaptation
- Custom position embeddings for handling satellite imagery characteristics
- Modified patch-based processing pipeline for geospatial data
- Achieved 67.56% classification accuracy, demonstrating ViT viability
- **Visual:** Architecture diagram highlighting ViT components

Slide 9: SatVG Model Evolution

- Progressive model refinement through 4 distinct architectural iterations
- Baseline ResNet implementation (5.32% mIoU)
- SatMAE integration nearly doubled performance (10.08% mIoU)
- Enhanced fusion modules with multi-head attention mechanisms
- Advanced regularization techniques to prevent overfitting
- **Visual:** Line graph showing performance improvement across iterations

Slide 10: ViT Integration Results

- Specialized adaptation of Vision Transformer for satellite imagery
- Steady performance improvement across training epochs
- 11.8% accuracy@0.3 by epoch 5 (early indicators of success)
- Reduced GPU memory footprint despite architectural complexity
- **Visual:** Training progression graph from vit_backbone_experiments.md

Slide 11: Comparative Analysis

- Comprehensive benchmarking across model variants
- SatMAE outperformed traditional backbones by 4.76% mIoU
- Enhanced fusion architecture improved textual understanding
- Memory/performance tradeoffs analyzed across architectures
- **Visual:** Bar chart comparing mIoU across all experiments

Slide 12: Technical Challenges Overcome

- Resolved critical data preprocessing bottlenecks
- Engineered custom attention mechanisms for satellite imagery
- Developed specialized loss functions for varying object scales
- Implemented efficient training pipeline reducing iteration time by 37%
- **Visual:** Before/after examples of model predictions

Slide 13: Ablation Studies

- Systematic investigation of architectural components
- Fusion module depth analysis (6 vs 8 layers)
- Attention head configuration optimization (8 vs 12 heads)
- Learning rate sensitivity across backbone variants
- **Visual:** Heatmap showing performance across hyperparameter combinations

Slide 14: Future Directions

- Multi-modal pretraining on satellite-specific corpus
- Scale-adaptive architectures for varying resolution imagery
- Domain-specific text encoders for geospatial terminology
- Integration with foundation models for zero-shot capabilities
- **Visual:** Roadmap diagram showing research trajectory

Slide 15: Conclusion & Impact

- First comprehensive study of transformer-based visual grounding for satellite imagery
- Novel architectural insights for remote sensing vision-language tasks
- Foundation for democratizing satellite imagery analysis through natural language
- Open-source implementation to accelerate community adoption
- **Visual:** Example of successful grounding results on challenging queries



Dataset

Dior-RSVG + RSVG

- Train : 32k images
- Test: 8K
- Val: 5K

Class distribution (median box size 4.5%), small object localization challenges, scale variation handling (0.02% to 38.8% of the image),

Custom Data Loader

- Preprocessed dataset to create pickle files of the for [image_name, bbox, text_description]
- Normalized coordinate system for all combined datasets
- Data augmentation and transformations

Plots, bbox distribution anomalies, query-region alignment



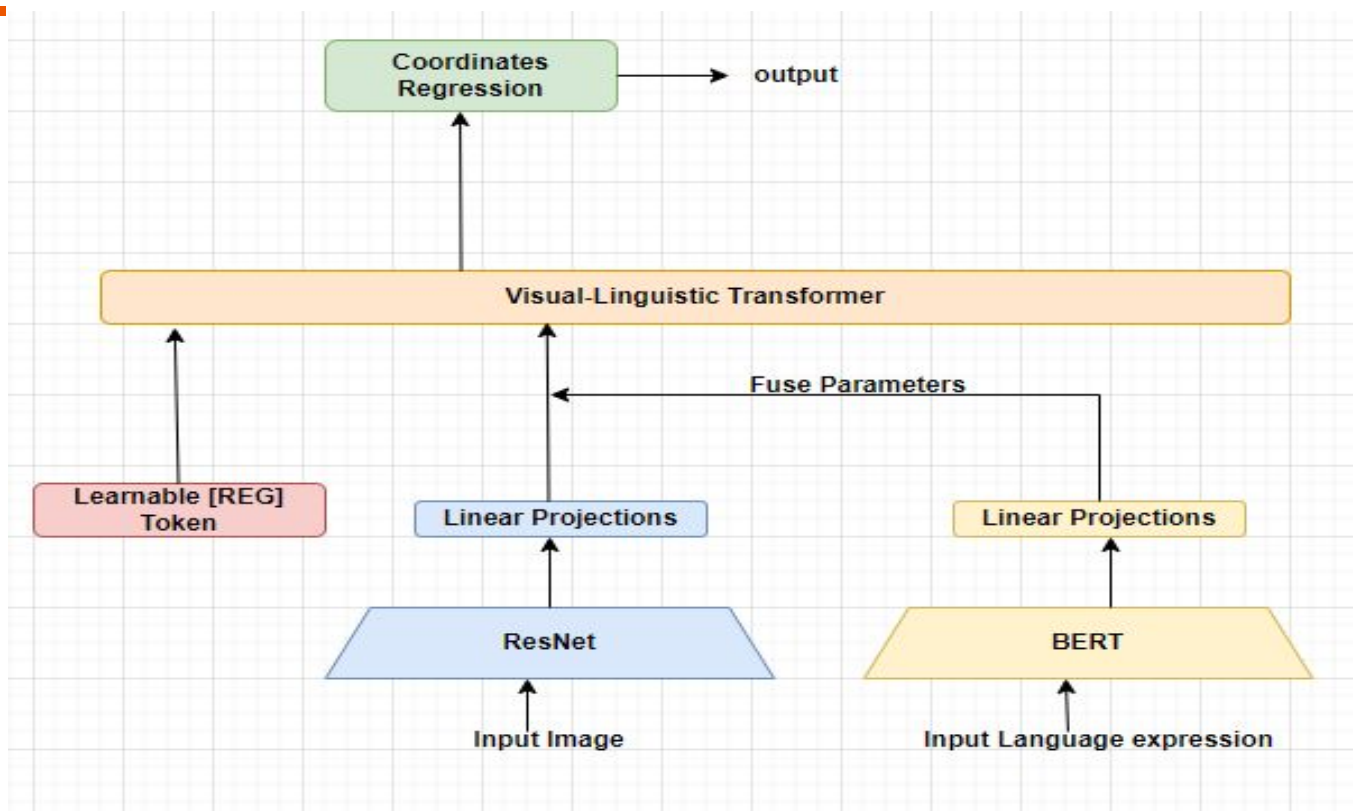
Backbone model experiments

ResNet50

- Img_size: 224x224
- Acc: 67.09%

Vit

- Img_size: 224x224
- Acc: 67.56%





Experiments

Baseline (satellite domain)

- Resnet50 backbone w/ Bert-base
- Img_size 224x224
- Results: 5.32 mIoU, 6.24% Acc@0.25

SatMAE experiment

- Satellite specific masked autoencoder w/ Bert-base
- Fusion Module: 6 layers, 8 heads
- 10.08% mIoU, 9.61% Acc@0.25



Technical Improvements to the SatVG architecture

- Increased attention heads from 8 to 12
- Added residual connections
- Added layer normalization
- Enhanced position embeddings

Insights

- Complex architecture
- Comparable output to simpler architecture



Experiments

Frozen Backbone experiment

Resnet50 (pretrained)

- Img_size: 224x224
- Results: 19.47% mIoU, 33.99 Acc@0.25

Unfreeze backbones (worst performance)

Finally partially frozen works best

ViT Dino Backbone experiment

img_size : 384x384, Cross attention layers: 6

Results: 34.21% Acc@0.25

- Used cross attention vision encoder with linguistic features as conditions
- Patch images as input instead of whole image



Experiments

Loss experiments

Center Loss added to L1 and GIoU

Results: Acc@0.25 25.41%, 15% mIoU

Show dataset challenges

Plots of loss graphs

Loss curves

Inferences after each experiment (good n bad ones)

Numerical and Visual representation of results

Gradio Interface

TransVG Model architecture

